Chemical Distribution, Blending and Supply Terminals Facility-Class Design Basis

The Third Facility Class of the Chemical Sector Design Basis

DBA-CHEM-DI-FC3 — Facility-Class Design Basis, DBA-CHEM Series

Issued Under: Chemical Sector Framework (DBA-CHEM_Sector_Framework v0.10); DBA-MA Parent Framework

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document sets out the design basis for chemical distribution: the terminals, repackagers, blenders and retailers that store chemicals and deliver them to the people who use them. It is the third facility class in the DBA-CHEM series. The *Chemical Sector Framework* (DBA-CHEM_Sector_Framework) states the sector's design basis in general terms, and this document applies it to the last link before a chemical reaches its user. It names the events such a facility must withstand, traces how each event reaches people, sets the parameters that size the protection, and states what the facility must be able to do on its own when outside help is cut off. It describes no particular site. A design basis for a specific facility starts from this document and adds only what is particular to it.

This class turns the production class around. A production plant is dangerous mainly to its neighbors. A distribution facility is often modest in what it could do to its neighbors and severe in what its loss would do to someone else, because it may be the only place a water utility or a farm region gets a chemical it cannot do without (framework §3.4). For many facilities in this class, the governing harm is losing the supply, not the release.

1.2 Scope

The class covers fixed facilities that receive chemicals in bulk, store them, and repackage, blend or sell them onward: bulk liquid terminals and tank farms, chlorine and bleach repackagers, blending and packaging plants, and agricultural chemical retailers. It covers their storage, their transfer equipment, their control and inventory systems, and the utilities they need. Production is a separate class, DBA-CHEM-PR-FC1 (*Chemical Production and Synthesis*), and so is transport, DBA-CHEM-TR-FC2 (*Bulk Chemical Transport*).

Three boundaries define the class.

With transport. Material belongs to DBA-CHEM-TR-FC2 until custody passes to the receiving facility, and to this class from that point (*Bulk Chemical Transport* §1.2). Material in transport custody that stands on the facility's property awaiting unloading remains the transport class's until custody passes, but it is counted in the facility's source term (Section 3.3), and the dependency is recorded at that boundary. At West a railcar of more than 100 tons of ammonium nitrate stood unloaded about 190 feet from the store and overturned without detonating (*West Fertilizer Explosion* F-5).

With the user. Material belongs to this class until custody passes to the user. A chlorine container or bleach delivery becomes part of a water treatment plant's inventory under DBA-W-FC1 (*Water Treatment Plant*) once it is received there. Anhydrous ammonia in a farmer's field tank belongs to the farm.

With ports and warehouses. A port warehouse holding chemicals in transit is a transportation node under DBA-TRAN-ND-FC1 (*Nodes and Terminals*). Following the rule the *Transportation*

Sector Framework states for tank cars (*Transportation Sector Framework* §1.5, §6.3), the harm a release or explosion of the stored chemical would do belongs to this class, and the harm to the port as a transport node belongs to Transportation. Both use the same release model. Explosive-capable and inhalation-hazard material held in a port, warehouse or customs area because it has been detained, seized, abandoned or refused, or because it is awaiting disposal, is within this class for Chain R whoever holds custody of it, and the custodian bears the facility's obligations under this document. Such material belongs to no facility's business, and a provision must place it: the cargo at Beirut was held by state authorities for nearly six years, received by no facility and claimed by no one (*West Fertilizer Explosion* F-11; Section 2.4).

On the author's decision of 5 October 2026, agricultural chemical retailers that hold ammonium nitrate or anhydrous ammonia belong to this class, because their governing harm is an explosion or release, which is the chemical chain. The Food and Agriculture distribution class, DBA-FA-DR-FC3 (*Storage, Distribution and Retail*), covers food storage and the cold chain and does not take them.

1.3 The Reference Configurations

Four kinds of facility are used as reference cases.

A chlorine and bleach repackager supplying water utilities is the governing configuration for supply loss. It receives chlorine in rail cars and refills it into ton containers and cylinders, or makes and delivers sodium hypochlorite solution. It may hold far less than a production plant. But it may also be the only supplier a group of water utilities can reach in the time their own stock lasts, and bleach cannot be stockpiled for long because it loses strength in storage. This configuration sets the supply-loss chain.

A bulk liquid terminal on a port or river is the governing configuration for release size. It holds large volumes of many chemicals in a tank farm, transfers them between ships, barges, rail cars, trucks and pipelines, and often sits in an industrial district near a city. This configuration sets the size of the release.

An agricultural chemical retailer holding ammonium nitrate and anhydrous ammonia is the governing configuration for explosion near a town. It is usually small, in or beside a rural town, and seasonal in its inventory. Ammonium nitrate stored in bulk can detonate when it is caught in a fire. This configuration sets the explosion chain, and the class's reference event is of this kind (Section 2.4).

A port, warehouse or customs store holding explosive-capable or inhalation-hazard material awaiting disposal is the configuration for material held in custody (Section 1.2). Its quantity is set by the cargo, not by any business of the custodian, and the time the material is held is a parameter of Section 3.3 (*West Fertilizer Explosion* F-11).

A requirement that names one configuration applies to that configuration. A requirement that names none applies to all of them.

1.4 Class Type

This is a physical facility class under the author's determination of 30 September 2026 (*Facility-Class Types Are Physical, Coordination and Logical*). The harm it can do comes mainly from equipment and the material it holds. In the framework's structure it sits under Distribution (DisCo), one of the sector's four function classes (framework §3.1, §3.4).

1.5 Relationship to Governing Documents

This instrument is issued under the *Chemical Sector Framework* and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, principles or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton, this instrument develops the class-level content ahead of it and records in Section 9 what the framework should take up. References of the form "framework §N" are to the *Chemical Sector Framework* in force.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty, the property of a protective function that executes independently of reachable digital authority, is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*Envelope*) reserves the term "envelope". Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

2. The Chemical Distribution Facility Class

2.1 Defining Characteristics

Three features define the class.

First, it is often critical to someone else. A distribution facility's importance is measured less by what it holds than by who depends on it. A small repackager can matter more to a city than a large terminal, if it is the city's only source of disinfectant.

Second, it stores without making. There is no reaction to control, so the dangers are in storage and transfer: a tank or warehouse that fails, a transfer that goes wrong, or a stored material that becomes dangerous when it is heated or contaminated. Ammonium nitrate is the extreme case. It is stable in normal storage, and it can detonate in a fire.

Third, it is lightly governed. Many facilities in the class fall below the thresholds of the main chemical-safety rules, or outside them altogether, and the federal chemical-facility security program has lapsed (framework §2.1). The rules that cover ammonium nitrate storage were called "a patchwork that has many large holes" by the Chemical Safety Board's chairperson, in testimony to the Senate Committee on Environment and Public Works on 27 June 2013, and "a patchwork of federal and state regulations with huge gaps" by its supervisory investigator, in a statement at Dallas in April 2014; the description is theirs, not the Board's final report's (Section 8.4). A design basis is often the only engineering statement these facilities will ever have of what they must withstand.

2.2 The Hazard-Class Sub-Index

The class is sub-indexed by hazard class under the framework's UN/DOT anchoring (framework §3.6), and by the UN number of what each facility delivers, which is the key to who depends on it (framework §6.1). Chlorine (UN 1017, Division 2.3) and sodium hypochlorite solution (UN 1791, Class 8) tie a facility to water treatment. Anhydrous ammonia (UN 1005) ties it to agriculture and to water treatment plants that use chloramine. Ammonium nitrate fertilizer (Division 5.1) ties it to agriculture, and it is the one material in the class that can detonate. The NFPA 704 diamond posted on each tank and building gives the framework its first, rough measure of how severe a substance is (framework §4.3). Where a facility holds anhydrous ammonia near ammonium nitrate, the two are ranked in its design basis and in its responders' plan by what each can do, not by which a regulation covers. At West the firefighters, drilled on the regulated ammonia, turned to cool its vessels inside the blast radius of the unregulated nitrate (*West Fertilizer Explosion* F-4; Section 4.2).

2.3 Deployment Typologies and Size Range

The class ranges from a large port terminal with hundreds of tanks to a farm-supply store with a fertilizer shed and a few ammonia tanks. Size does not set importance. A small facility with a large stock of a dangerous material near homes, or a small facility that is the only supplier to a critical user, can matter as much as a large one, because what counts is how severe the harm could be (framework §4.1).

2.4 Consequence Context

The class has two governing harms, and either can govern a given facility. The first is a release or explosion that reaches the people around the facility. The second is a supply loss that reaches the people who depend on the facility's customers, such as a city whose water utility runs out of disinfectant.

The governing reference event is the explosion at the West Fertilizer Company in West, Texas, on 17 April 2013 (DBA-CHEM-RE-5). A fire first reported at about 7:29 p.m. reached fertilizer-grade ammonium nitrate held in plywood bins in a wood-framed building without detection or sprinklers, and at about 7:51 p.m. about 30 tons of the 40 to 60 tons stored detonated, on the

account of the U.S. Chemical Safety and Hazard Investigation Board. Fifteen people died, twelve of them emergency responders, eight of those volunteer firefighters, and more than 260 were injured. Two schools, a nursing home and apartments nearby were destroyed or demolished. The Board found the Texas certification curriculum silent on the explosion hazard of fertilizer-grade ammonium nitrate, and the guidance available to responders vague, inconsistent and oriented to transport, although the 2012 Emergency Response Guidebook and one producer's data sheet warned that the material may explode or detonate. It found that 19 of the 40 Texas facilities holding more than 10,000 pounds of it, 48 per cent, were within half a mile of a school, hospital or nursing home; the 1,351 facilities it counted nationally are those above the anti-terrorism screening thresholds, for which it gave no proximity figure. The facility had reported its ammonium nitrate to the fire department and the planning committee in 2012. What responders and neighbors did not know was that it could explode.

The Board did not determine what started the fire. The Bureau of Alcohol, Tobacco, Firearms and Explosives determined in May 2016 that the fire was intentionally set; that determination is the Bureau's, its investigation remains open, and this class makes no finding on it. The class's requirements for a fire at such a store do not depend on its cause (Section 5.1; *West Fertilizer Explosion* F-6). The design-basis reading is that a small facility in a small town held enough material to destroy part of the town, and that its responders and neighbors, told what it held, were not told what it could do. The DBA Reference Event Register names Emergency Services, Government Facilities (education facilities), Healthcare and Public Health, and Water and Wastewater Systems as the sectors the event impacted.

The event was proposed by this class and conferred by the framework at its §9.1 as DBA-CHEM-RE-5, on the author's decision of 6 October 2026. Its instrument, *The West Fertilizer Explosion, April 2013*, is written and, under the determination of 9 October 2026 that gives reference events their own project, is held by Project 11. This revision takes in what it returns, and the account above is conformed to it (Section 10.7).

The explosion at the port of Beirut on 4 August 2020 was proposed here as the upper bound of the explosion chain and conferred by the framework as DBA-CHEM-RE-6. On the author's triage ruling of 9 October 2026 it is withdrawn as a reference event and its designator is retired; its two findings, *Beirut Port Explosion* F-1 and F-2, are carried by *West Fertilizer Explosion* F-11 and F-12, and it is cited here as an illustration. About 2,750 tonnes of ammonium nitrate, on the Lebanese government's account, arrived at Beirut in November 2013 aboard a ship later abandoned, were moved into a port warehouse under a judge's order in October 2014, and stayed there until they detonated on 4 August 2020. On Human Rights Watch's reading of official documents, many authorities held part of the decision on the cargo and none held the whole, and at least six requests to re-export or sell it were returned, referred or declined; Human Rights Watch's conclusions are its own, and no finding of fault is made here. Beirut shows what happens when a dangerous material sits in custody for years, belonging to no one's design basis (Sections 1.2 and 7.4; *West Fertilizer Explosion* F-11, F-12).

3. The Facility-Class Design Envelope

3.1 Why Distribution Is the Third Class

The framework puts distribution third because it is where the sector's importance to other sectors concentrates, and where the stricter supply criteria for dependent facilities will apply most often (framework §3.4, §8.2). This document takes that ordering as given.

3.2 The Two Consequence Chains

Chain R — release or explosion. A failure, fire or deliberate act releases or detonates stored material, and the harm reaches the people around the facility through the air or through blast. It runs in minutes or seconds. Its size is set by the shared release model of Section 4, which this class extends to explosion.

Chain S — supply loss. The facility stops delivering a chemical that a dependent facility cannot do without, for longer than the dependent facility's stock lasts. The harm reaches people through the dependent facility's own consequence, such as loss of disinfection at a water treatment plant or loss of fertilizer at planting time. It runs in days.

In the production class Chain R governs and Chain S is secondary. In this class either may govern, and the design basis is stated for both. A strike on a facility can open both at once, and for the repackager that serves a group of water utilities, Chain S is the reason it is a target at all.

3.3 Design Parameters at Class Resolution

Five parameters govern the envelope. The first is the governing source term: the most material a single credible event can release or detonate, which for ammonium nitrate is the largest mass that a single fire can involve. The second is the separation distance: how far the facility is from schools, hospitals, nursing homes and homes, against the distance at which its release or blast does serious harm. The third is the dependent facility's stock, in days, which for bleach is limited by how long bleach keeps. The fourth is the replacement time: how long it would take a dependent facility to find and receive the same chemical from another supplier. The fifth is the supply-continuity margin, which is the third parameter minus the fourth. When it is negative, the dependent facility runs out before it can be resupplied, and Chain S governs.

The governing source term for detonation is the largest mass a single fire can involve. Its yield uncertainty is stated and is not used to reduce it, and it includes every mass on site that a detonation could initiate, material in transport custody awaiting unloading among it (Section 1.2). At West about 30 tons detonated of 40 to 60 stored, how much burned first is unknown, and a railcar of more than 100 tons stood unloaded about 190 feet away (*West Fertilizer Explosion* F-5).

The separation distance is held by two parties: the facility, which holds the quantity, and the authority that permits building near it, which holds the distance. Built among fields in 1962, the facility at West came to stand within about 450 to 1,300 feet of apartments, a nursing home and

two schools built after it (*West Fertilizer Explosion* F-2). A new school, hospital, nursing home or dwelling within the distance is a relaxation under Section 6.8. Occupancy within the distance is counted at the worst hour: the schools at West were empty at the hour of the detonation, and on a school day, the Board noted, about 1,486 students would have been in the four schools nearby (*West Fertilizer Explosion* F-5).

For material held awaiting disposal (Section 1.2) a sixth parameter applies, the time held: how long the material may stay in custody before its staying needs a recorded basis under Section 6.8 (*West Fertilizer Explosion* F-11).

This document states the parameters and the reasoning behind them. Their values depend on facilities, customers and supply chains, and are not given here as numbers (Section 10.1).

4. Use of the Shared Chemical-Release Consequence Model

4.1 The Model Cited

The release model this class uses is the shared chemical-release consequence model, held for now in *Chemical Production and Synthesis* §4 on the author's decision of 5 October 2026 (framework §6.3). This document cites it there and does not restate it. It is the model's second citer, after *Bulk Chemical Transport*, and it is owed notice when the model moves (Section 10.4).

4.2 What Distribution Adds

The model has four stages: how much is released, how it spreads, the concentration at which it harms people, and who is exposed. Distribution adds four conditions.

First, explosion. The model as written is built around toxic and flammable releases. Stored ammonium nitrate adds a third kind of harm: detonation, with blast overpressure and thrown debris. For that material the model's spread and endpoint stages are replaced by the distance at which blast overpressure causes serious injury and destroys buildings, and the exposure stage counts everyone inside that distance. This addition is made here and offered back to the model's home (Section 10.4). The exposure stage also counts fragments, glazing, post-blast fire and the infrastructure the blast reaches, water among it, and counts occupancy at the worst hour (Section 3.3). At West a foundation fragment struck the nursing home, glazing injured people indoors, fires followed and the city's water system failed (*West Fertilizer Explosion* F-5).

Second, mixed storage. A terminal holds many chemicals in one place, and a fire in one tank can heat, breach or ignite its neighbors. The amount released is counted for everything a single fire can involve.

Third, the responders are among the exposed. At West twelve of the fifteen dead were responders. The exposure stage counts the people who will come toward the facility, not only those who live near it, and the design basis requires that they know what is stored there and how it behaves before they arrive. Knowing in advance means a pre-incident plan made by the facility with the fire authority and exercised, not a report filed. The plan reaches every party that will

respond, mutual aid included, and carries the decision to withdraw to each on arrival. At West the ammonium nitrate had been reported in 2012 and no plan followed, and responders from four departments came with no incident command and no public alert (*West Fertilizer Explosion* F-3; Section 5.1).

Fourth, the combined chain. A detonation can initiate a release from a nearby inhalation-hazard store, and the exposure counts both. At West two ammonia vessels, 34,000 pounds on the company's report, stood about 150 feet from the fire; the blast damaged them, one with its relief valves capped, and a valve leaked, a near-miss the Board found a siting analysis would likely have foreseen (*West Fertilizer Explosion* F-4). The combined chain is offered with the detonation addition to the model's home (Section 10.4).

5. Design Basis Events: Non-Adversarial Envelope

The framework lists the kinds of event a chemical facility must withstand but does not yet develop them (framework §4.2). This section develops each for a distribution facility, using the series' four headings: the event, the harm that makes it count, what the facility must do, and for how long. Durations are given as class parameters until the framework sets values (Section 10.2).

5.1 Fire Involving Stored Explosive-Capable Material

Postulated event a fire in or near storage of ammonium nitrate or another material that can detonate or react violently when heated. The event is postulated whatever its cause, accidental or set (*West Fertilizer Explosion* F-6).
Consequence threshold detonation, or a violent reaction, whose blast or release reaches people off site or reaches the responders.
Required function such material must be stored so that a credible fire cannot involve enough of it to detonate with serious off-site harm: in buildings that do not burn, away from fuels and contaminants, divided into separated quantities, and with fire detection and suppression that do not depend on anyone being present. The facility must be far enough from schools, hospitals and homes for the quantity it holds, or it must hold less. Responders must know in advance what is stored and when not to approach.
What makes a store detonable in a burning building is part of the requirement. The products of the building's own fire are contaminants, so that a store is noncombustible only if what is above and around it is; ventilation and the confinement of a bin are conditions of the requirement; separated quantities are separated in fact, with no openings between bins; sources of ignition, electrical service among them, are kept out of the store or protected; and the suppression holds its own water or states the supply it depends on. At West the Board found that combustible construction and nearby combustibles contributed to the fire and likely the detonation, that the likely mechanism was contamination of the pile by soot, molten asphalt and polyvinyl chloride from the burning structure under limited and then increased ventilation, that the main bin opened into its neighbor, and that the only fire water was a hydrant about 1,600 feet away (*West Fertilizer Explosion* F-1).
The separation distance is held with the authority that permits building near the facility (Section 3.3), and a new school, hospital, nursing home or dwelling within it is a relaxation requiring a recorded basis or a smaller quantity (Section 6.8; *West Fertilizer Explosion* F-2). An inhalation-

hazard store near explosive-capable material is separated from it by a distance sized against blast, and the protective devices of that neighboring store are kept in service (*West Fertilizer Explosion* F-4).
The decision not to fight the fire, and the distance to which responders withdraw, are set in the pre-incident plan of Section 4.2 and tested against the minutes in which such detonations occur; at West the detonation came about twenty minutes after the first report. A public alert to the people within the blast distance is a required function at the first report of fire, under framework §4.3 (*West Fertilizer Explosion* F-3). The detection and suppression credited under this section meet the independence of Section 5.6 against a set fire, so that no single act by a person on site disables them, the fire being one a person can set (*West Fertilizer Explosion* F-6).
Design-basis duration event-specific; the decision not to fight such a fire must be made before it is fought.

5.2 Release from Storage or Transfer

Postulated event failure of a tank, container or transfer connection holding an inhalation-hazard or flammable material, including during unloading from or loading into transport.
Consequence threshold a release whose footprint reaches people off site.
Required function storage must be divided and contained so that a single credible event does not release the governing source term entirely. Transfers must stop and isolate on their own when a hose, arm or connection fails, without waiting for a person or a networked command.
Design-basis duration event-specific.

5.3 Cyber Attack on Terminal Control and Inventory Systems

Postulated event intrusion into the systems that control tank levels, valves and transfers, or that record what is stored and what is owed to customers, sending false commands or false records.
Consequence threshold a tank overfilled or a transfer misdirected, a protective function defeated, or a dependent customer’s supply lost because the records on which deliveries are planned are false.
Required function overfill and transfer protections must not depend on the system that may be compromised. No change to the settings of a protective function may be made from any system an attacker could reach, except by a deliberate, authenticated and recorded act of a qualified person that no automated system can perform or finish alone. FD-LS (*Local Sovereignty*) calls this a governed act, and the term is used in that sense throughout. The facility must be able to confirm the stock it holds for its critical customers by a means independent of its business systems.
Design-basis duration indefinite; the condition lasts until the compromise is found and removed.
The second half of this category is particular to the class. A distribution facility can fail its dependents without anything physical going wrong, if the records that plan its deliveries are corrupted. A ransomware attack that takes down a repackager’s ordering and dispatch systems can cut a water utility’s supply as surely as a fire.

5.4 Loss of Power and Supporting Services

Postulated event loss of electric power, communications or other services on which storage, transfer, protection or delivery depends.

Consequence threshold a protective function lost, or deliveries to critical customers stopped beyond their stock.
Required function every protective function must reach its safe state on loss of power. Deliveries to critical customers must be possible without the lost service for the class duration.
Design-basis duration until the service is restored, for a class duration awaiting the framework (Section 10.2).

5.5 Loss of Incoming Supply

Postulated event loss of the facility's own supply, from a production plant outage or a cut transport route.
Consequence threshold the facility unable to supply its critical customers before their stock runs out.
Required function a facility that is the only supplier to a critical customer must hold stock or a second source sized against that customer's supply-continuity margin.
Design-basis duration the replacement time of Section 3.3.

5.6 Physical Attack

Postulated event deliberate attack on storage, on transfer equipment, or on the facility as a supplier.
Consequence threshold the release or explosion footprint reaching people, or the facility's critical customers losing supply.
Required function storage of inhalation-hazard and explosive-capable material must be protected against deliberate attack. Where protection is provided by more than one means, the means are credited against deliberate attack only when they are robust, diverse and independent: two or more, differing in kind, and sharing no location, power supply, control system, vendor or update path, so that no single act defeats them all (*Defense in Depth* §3.4). Where the facility cannot show independent protection against every combination of losses that would defeat it, the gap must be stated openly as a declared residual, in the form *Defense in Depth* §3.5 sets out (Section 6.7). The detection and suppression credited against a fire at explosive-capable storage are among the means so credited, and the fire is postulated as one a person on site can set (Section 5.1; *West Fertilizer Explosion* F-6).
Design-basis duration event-specific.

5.7 Natural-Hazard Exceedance

Postulated event flood, hurricane, earthquake or extreme heat beyond the design of storage or buildings.
Consequence threshold a release or explosion, or the loss of supply to critical customers.
Required function storage and its protections must remain within their design basis under the event. Material that becomes dangerous when wet, heated or mixed must not be stored where a foreseeable flood or heat can reach it.
Design-basis duration as in Section 5.4 where the hazard removes a service; event-specific otherwise.

Table 5-1. Facility-class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Framework source
Fire involving explosive-capable material	Fire, whatever its cause, reaches stored ammonium nitrate or similar	Chain R (blast)	developed here
Release from storage or transfer	Tank, container or connection failure	Chain R	§4.2 candidate (release from storage)
Cyber attack on control and inventory	False commands or false records	Chain R + Chain S	§4.2 candidate (control-system compromise)
Loss of power and services	Power or communications loss	Chain R + Chain S	developed here
Loss of incoming supply	Production outage or cut route	Chain S	§4.2 candidate (supply-chain denial)
Physical attack	Attack on storage, transfer or the supplier role	Chain R + Chain S	developed here
Natural-hazard exceedance	Hazard beyond design	Chain R + Chain S	developed here

6. Military-Action Extension: DBA-MA for Chemical Distribution

6.1 MA Extension Applicability Determination

Every facility makes the applicability determination the series makes mandatory (framework §7). The primary factor is exposure to military action, and the secondary factor is strategic value. For this class, strategic value is driven mostly by who depends on the facility. A repackager that is the only disinfectant supply for a metropolitan water system carries strategic value whatever its size (framework §7). So does a port terminal whose loss would cut a region's supply of a critical chemical, and a store of explosive-capable material near a strategic site. The determination is recorded, re-made when the defined geopolitical indicators change, and any scope reduction carries a recorded basis (Section 6.8).

6.2 The Four Characteristics Applied to the Class

Intentionality: the adversary selects the facility for whom it supplies, or for what its stored material could do to a town. Persistence: the adversary sits inside the facility's business and control systems long enough to learn its customers and its stock. Precision: the adversary chooses the moment a dependent utility's stock is lowest, or the fire that will reach the explosive-capable store. Denial: the adversary corrupts the records that would show the supply failing, and times the act to a moment when other suppliers are also short. The standard this class is tested against is a state actor who has found that the quickest way to cut a city's water is not to attack the water plant but to cut off its chlorine. Against an explosive-capable store the adversary may also rely on a set fire to draw responders and onlookers toward it in the minutes before it detonates, and on the hour to fill the schools nearby (*West Fertilizer Explosion* F-7).

6.3 The Governing MA Scenario

The governing military-action scenario for this class is supply denial, not release. An adversary takes the only disinfectant supplier for a group of water utilities out of service — by cyber attack on its ordering and dispatch systems, by physical attack, or both — at a time when the utilities' stock is low and other suppliers are stretched. No release occurs. Within days the utilities cannot disinfect, and the harm reaches the population through the water. The extension therefore requires that a critical supplier know who depends on it and what their stock is, that it hold stock or a second source sized against the worst of their margins, and that its ability to deliver not depend on systems an attacker can take down.

The release scenario remains: a precision strike on an inhalation-hazard store, or a fire set at an ammonium nitrate store. Against it the requirements of Sections 5.1, 5.2 and 5.6 apply. The set fire is stated with its delayed detonation: a fire set at the store draws responders and onlookers toward it, and the detonation reaches those who respond. The constructed analog of *West Fertilizer Explosion* §5.6 is the reference for its direction, and it is not a claim that the event at West was such an act (*West Fertilizer Explosion* F-7).

6.4 Hazard Labels as a Targeting Aid

The framework finds that hazard labels which warn responders also tell an attacker what to hit (framework §7.1). At a distribution facility the labels are the NFPA 704 diamonds on its tanks and buildings. The lesson of West runs the other way: the facility had reported its ammonium nitrate, but responders and neighbors did not know that it could explode, and paid for it. The design basis therefore keeps the labels and adds to them. Responders must have the facility's storage and hazard information in advance, and the design basis assumes the adversary has it too. The information reaches responders through the exercised pre-incident plan of Section 4.2, and with it the decision to withdraw (*West Fertilizer Explosion* F-3).

6.5 Cross-Sector Interaction

Under military action the class is tied to other sectors chiefly through its customers. Going out, its loss cuts disinfectant to water utilities, fertilizer to farms, and other chemicals to the sectors that use them. Coming in, it depends on production and transport for its own supply, and on power and communications to store safely and deliver. An attacker who cannot reach a well-protected water plant can reach it through its supplier, and an attacker who cannot reach a production plant can reach its customers through the terminal they share.

6.6 The Adequacy Result: Autonomous Safe Terminal Condition

The test of whether a facility's protection is adequate is whether it reaches what this document calls the Autonomous Safe Terminal Condition. When control and outside help are lost, the facility's own protection holds its stored material safely: transfers stop and isolate, tanks hold, and explosive-capable material stays separated, cool and unburned. Where that condition rests on

suppression, the suppression holds its own water or states the supply it depends on (*West Fertilizer Explosion* F-1).

The protective functions on which this condition rests are to be locally sovereign as FD-LS defines the property (*Local Sovereignty* §3.2). The decision is a measurement against a setpoint; the function is sited with the material it protects; no system an attacker could reach can stop, delay or trigger its action; and no such system can alter its setpoints, logic or bypass state except through a governed act. *Local Sovereignty* governs and is not restated here. An overfill valve that closes on the level it measures, and a transfer valve that closes when its hose separates, are the class's clearest cases. Local sovereignty describes what protects the facility when control and authorization have failed. It does not decide whether any action needs human authorization (*Local Sovereignty* §4).

The condition must also hold for a set period with no outside power, communications or support. That requirement is written here in this document's own words, and for the reason given in Section 10.5 it is not borrowed from the DBA-MA parent framework's local sovereignty principle.

For this class the safe condition has a second half. A facility that is a critical supplier must also be able to keep delivering to its critical customers for a set period without its business systems: it must know, from records it holds independently, who they are, what they need and how to reach them. The means of that second half are stated with it: the records, the people who can work from them, the stock that bridges the time to restore the business systems, and procedures exercised in advance. The facility's channels to its critical customers, its workforce and the public are prepared in advance and do not depend on its enterprise email, directory or web systems, and notice of a cyber attack on those systems goes to law enforcement and the national cyber authority in time for them to act. These elements are noted to the class by *LockerGoga Attacks* F-3, whose R-3 now carries R-6: Momentive processed orders, shipping, payments and production scheduling by hand with delays in sales, and Norsk Hydro, read as a cross-sector proxy, worked from paper, used stock to keep deliveries going and took back retired employees who knew the paper systems (*Chemical Enterprise Coordination and Control* §4.2, §4.7).

6.7 Adequacy and the Declared Residual

Against accidents, the safe terminal condition is achievable with the Section 5 requirements. Against a deliberate attacker, adequacy has limits. Two residuals are declared, in the form *Defense in Depth* §3.5 sets out. First, a large enough store of ammonium nitrate or of an inhalation hazard near people cannot be made safe against a determined attack by protection alone. The residual is managed by how much is stored and how far it is from people, which is why Section 5.1 allows the facility to hold less as well as to be farther away. Second, a sole supplier cannot be made immune to supply denial. The residual is managed by the dependent facility's own stock and by a second source, which is why Section 8.1 places part of the burden on the water utility's side. Against the set fire of Section 6.3, the decision not to fight it and the

public alert of Section 5.1 are the only protection for responders and the public, and the first residual's quantity and distance are stated against occupancy at the worst hour (Section 3.3; *West Fertilizer Explosion* F-7).

6.8 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter — a larger quantity stored near people, a shorter replacement time assumed, a dependent customer treated as having another source — must be justified against the class baseline, and the larger the relaxation the stronger the analysis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework does not yet state this principle, and Section 9.1 records it for the framework to adopt. The check applies to three relaxations the class's event names. A new school, hospital, nursing home or dwelling built within the separation distance is a relaxation, requiring a recorded basis or a smaller quantity (*West Fertilizer Explosion* F-2). Material held awaiting disposal beyond the time held of Section 3.3 needs a recorded basis for its staying, given by the holder of the disposition decision (Section 7.4; *West Fertilizer Explosion* F-11, F-12). And the check applies to those who hold a small facility's design basis with it (Section 9.2) as it applies to the facility (*West Fertilizer Explosion* F-9).

7. The Dual Design Basis, Stated in Full

7.1 The Two Design Bases

A distribution facility carries two design bases at once: a physical design basis, defined in this document, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). They meet at the point developed in Section 7.4.

7.2 The Physical Design Basis

The physical design basis is the protection that keeps stored material safe without depending on any networked system: buildings and containment that do not burn or fail, separation of quantities and of incompatible materials, overfill and transfer shutoffs, fire detection and suppression, and separation distance from people. Much of it is passive, which is the class's strength.

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Overfill and transfer shutoff logic that behaves predictably within a stated range of inputs, and that has been formally tested as installed, is exempt from the Command Broker requirement for the actions it takes while its inputs stay in that range (*Layer Decomposition and Determinism* §4.3). It is not thereby below the line; it is exempt because its behavior is bounded by proof (*The Bright Line* §4.5). A component that has not been shown to meet this test is covered by the rules, and each of its actions is judged on its consequence.

7.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems in distribution are mostly found in planning: demand forecasting, inventory optimization, delivery scheduling and routing, and the automated ordering systems that connect a supplier to its customers. They are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*). CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3).

The worked example of this class is the inventory system that is covered by the rules. Inventory optimization exists to run stock down, because stock costs money. For most chemicals that is harmless. For a sole supplier's stock of disinfectant held for water utilities, each decision that runs that stock below what the utilities need is an action judged on its consequence. That consequence is the utilities' loss of disinfection if anything then goes wrong, and on that consequence the decision is above the line and needs Command Broker authorization. The same system deciding stock levels for a chemical no critical customer depends on is judged on that lesser consequence, and the decision may fall below the line. The difference is not in the system. It is in who depends on the stock.

7.4 The Interaction Point: The Critical Customer's Margin

The two design bases meet at the supply-continuity margin of Section 3.3. The control and planning systems must be architecturally incapable of executing an above-line action originating from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization, and that includes any action that would take a critical customer's margin below zero. The person who authorizes such an action must see the margin, which means the facility must know its critical customers' stock, and the information must be trustworthy under the integrity-of-basis requirement of *Indication Integrity*.

The class's other interaction point is the disposition of material held awaiting disposal (Section 1.2). Each decision on its disposition is an action judged on its consequence, and it has a single named holder who can authorize or refuse it. It may be discharged in advance, under FD-BL-D1 (*Mode of Discharge*), by a standing rule that the material is removed or destroyed within the time held of Section 3.3 unless the holder authorizes, on a recorded basis, that it stay; and the basis, including the warning of what the material can do, reaches the holder intact under FD-BL-D2 (*Indication Integrity*). At Beirut many authorities held part of the decision and none the whole, and on Human Rights Watch's reading the report that reached the executive in July 2020 omitted a warning of a catastrophic explosion at the port (*West Fertilizer Explosion* F-12). Where the custodian is a state, its holder is outside Chemical Enterprise Coordination and Control, and the dependency is noted to that class and to the framework (Section 10.8).

7.5 Scope Boundary

The physical design basis is defined in full in this document. The AI-governance design basis is named and referenced, not redefined; its governing text is *The Bright Line* with its determinations, and the Command Broker family under its register. Where the two are cited together, the physical requirement governs the facility's protection and the AI-governance requirement governs the authorization of any system that proposes an action affecting it.

8. Cross-Sector Dependencies and Regulatory Context

8.1 What Other Sectors Depend On

This is the class in which the framework expects the forward supply rule to apply most often (framework §3.4, §6.1). When a distribution facility is the only supply for a facility in another series that is rated Tier 2 or Tier 3, it must meet stricter criteria for that supply. The relationship is identified by the chemical's UN number (framework §6.1). Chlorine (UN 1017) and sodium hypochlorite (UN 1791) tie the class to DBA-W-FC1. Anhydrous ammonia (UN 1005) and ammonium nitrate tie it to DBA-FA-AG-FC1 (*Agricultural Production*), and ammonia also ties it to DBA-W-FC1 where utilities use chloramine.

The burden of a sole supply relationship should not fall on the supplier alone. A water utility that depends on one supplier for disinfectant carries part of the risk in its own stock and its own second source. The water series already requires an on-site chemical inventory and an alternate disinfection pathway (*Water Treatment Plant* §4.5). This document places the matching obligation on the supplier side, and who bears the cost of the stricter criteria is the same question Project 9 OI-17 asks for cooling water (Section 10.6).

8.2 What the Class Depends On

The class depends on production plants and on transport for its supply, on electric power to store and transfer safely, and on communications and business systems to plan and dispatch deliveries. It depends on fire water to suppress a fire at its store, and states the supply it depends on where it does not hold its own; at West the nearest hydrant was about 1,600 feet away and the hose laid from it fell 700 feet short (*West Fertilizer Explosion* F-1).

8.3 Boundaries

The three boundaries of Section 1.2 are restated here as the class's cross-series interfaces. With transport and with the user, the boundary is custody. With the Transportation series at ports and warehouses, the release or explosion is this class's and the harm to the transport node is Transportation's. Agricultural chemical retailers are this class's, not Food and Agriculture's. Material held awaiting disposal at a port or warehouse is this class's for Chain R whoever holds custody (Section 1.2), and DBA-TRAN-ND-FC1 (*Nodes and Terminals*) records the point at its §4.4 (*West Fertilizer Explosion* F-11). Material in transport custody awaiting unloading on the

facility's property is counted in the facility's source term, with the dependency recorded at the custody boundary of *Bulk Chemical Transport* §1.2 (*West Fertilizer Explosion* F-5).

8.4 Regulatory and Institutional Context

The class is governed in part by the Environmental Protection Agency's Risk Management Program and the Occupational Safety and Health Administration's Process Safety Management standard, for facilities that hold covered chemicals above the threshold quantities. Neither covers fertilizer-grade ammonium nitrate. The federal standard that did apply to its storage at West, 29 CFR 1910.109(i), the Board found little known, rarely enforced at fertilizer facilities and permissive of wooden bins. The county and state applied the agricultural use exemption of the Emergency Planning and Community Right-to-Know Act to the facility, so that it was not in the county's hazardous-materials plan. The retail exemption from Process Safety Management kept the facility's ammonia outside that standard; the Occupational Safety and Health Administration narrowed the exemption in July 2015, and the narrowing was vacated in September 2016. The facility had not filed a Top-Screen under the federal chemical-facility security program, and no fire code applied to it. The description of these rules as a patchwork is the Board's chairperson's, in his testimony to the Senate Committee on Environment and Public Works of 27 June 2013, and its supervisory investigator's, at Dallas in April 2014, not the final report's (Section 2.1). Storage is otherwise governed by state and local fire codes where they apply, and transfers to and from transport by the hazardous materials regulations (49 CFR Parts 171 to 180). The federal chemical-facility security program lapsed in July 2023 (framework §2.1); its status is to be confirmed at each revision of this instrument.

Executive Order 13650, *Improving Chemical Facility Safety and Security*, issued on 1 August 2013 after West, directed at its section 6(b) the development of a list of potential regulatory and legislative proposals to improve the safe and secure storage, handling and sale of ammonium nitrate; it did not itself improve the regulation. What followed was partial. NFPA 400 (2016 edition) prohibits combustible construction for new ammonium nitrate storage where it is adopted. Texas adopted rules in 2014 and a statute in 2015. The Board recommended that the Environmental Protection Agency list fertilizer-grade ammonium nitrate under the Risk Management Program (its R3) and that the Occupational Safety and Health Administration cover it under process safety management or revise its explosives standard (its R5); R3 was classified closed with unacceptable action in March 2019 and reopened in March 2023, and R5 is shown as open, on the record the reference-event instrument holds. The Agency's amendments of 2017 did not add ammonium nitrate, and the Agency cited the Bureau's arson finding when it delayed them; it rescinded most of them in 2019, named ammonium nitrate a priority chemical for review of its list in 2022, and in February 2026 proposed revising the Safer Communities by Chemical Accident Prevention rule it had issued in 2024. No federal rule listing ammonium nitrate for accident prevention had been found at 8 October 2026. The class's requirements do not wait on that listing. The Board's findings at West are the class's practice reference for Sections 4.2, 5.1

and 6.4, and who holds a small facility's design basis with it is stated at Section 9.2 (*West Fertilizer Explosion* F-8, F-9).

9. Inheritance Map and Anticipated Derivatives

9.1 The Inheritance Map

The series requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines itself. Table 9-1 is that map for this class. Where the framework is a skeleton, the class content is developed ahead of it and is marked for the framework to take up.

Table 9-1. Inheritance map for DBA-CHEM-DI-FC3.

Element	Source	Inheritance mode
Function-primary placement, DisCo function class	framework §1.3, §3.1, §3.4	Inherited unchanged
Hazard-class sub-index and UN-number linkage	framework §3.6, §6.1	Inherited unchanged, applied to the reference configurations
Design-basis principles (consequence severity, not likelihood)	framework §4.1	Inherited unchanged
Event categories	framework §4.2 (stubs)	Developed here ahead of the framework; for the framework to take up
Bright Line and Command Broker architecture	framework §5; FD-BL; Command Broker family	Inherited unchanged, referenced
Forward supply rule	framework §3.4, §6.1	Inherited; this is its principal home
Shared chemical-release consequence model	*Chemical Production and Synthesis* §4	Cited; explosion added (Section 4.2), offered back to the model
Boundary with the Transportation series at ports	*Transportation Sector Framework* §1.5, §6.3	Applied by analogy to stored material
MA applicability and hazard labels	framework §7, §7.1	Inherited with adaptation
Local sovereignty of protective functions	FD-LS §3.2	Cited directly
Independent protection and declared residual	DBA-MA-DiD §3.4, §3.5	Cited directly; for the framework to take up
Consequence-basis conditionality check	not stated in the framework	Stated here; for the framework to adopt
Reference event DBA-CHEM-RE-5	framework §9.1	Conferred; instrument written, held by Project 11; taken in (Sections 2.4, 10.7)
Findings of *West Fertilizer Explosion* F-1 to F-7, F-9, F-11 and F-12	*West Fertilizer Explosion*	Taken in (Sections 1.2, 1.3, 3.3, 4.2, 5.1, 6.2 to 6.4, 6.6 to 6.8, 7.4, 9.2); elements for others noted (Section 10.8)

Element	Source	Inheritance mode
Consequence chains, design parameters and the Autonomous Safe Terminal Condition	—	Defined here

9.2 Anticipated Derivatives

Site-specific design bases inherit this class document by reference and add only what is particular to them, each carrying a recorded applicability determination for any scope it reduces. A facility that cannot hold a site-specific design basis of its own, as an agricultural retailer usually cannot, holds it with others: with the fire authority, through the pre-incident plan of Section 4.2; with its insurer, through loss-control surveys that address the explosion hazard; and with its supplier, through stewardship made a condition of sale. The consequence-basis conditionality check of Section 6.8 applies to each of them as it applies to the facility. No external party held a design basis for the facility at West, and after the event one producer made a certification of compliance a condition of sale, which the Board judged could give reasonable assurance (*West Fertilizer Explosion* F-9). The supplier's part is noted to the producers' classes (Section 10.8). A configuration-specific instrument for ammonium nitrate storage is expected beneath this class, and its designator is the framework's to confer. The class proposed two reference events, and the framework conferred them at its §9.1: the explosion at West as DBA-CHEM-RE-5, whose instrument is written and held by Project 11, and the explosion at the port of Beirut as DBA-CHEM-RE-6, withdrawn on the author's triage ruling of 9 October 2026 with its designator retired, its two findings carried by *West Fertilizer Explosion* F-11 and F-12 (Section 2.4). The use case tracing the supply-denial scenario of Section 6.3 through to a water utility's loss of disinfection is written as DBA-CHEM-UC-3 (*The Sole Disinfectant Supplier Taken Down*), and it pairs with the water series' own use cases.

10. Gaps and Open Items

10.1 Quantitative Design Parameters

The governing source term, separation distance, dependent stock, replacement time and supply-continuity margin of Section 3.3 are stated as parameters, not numbers. Their values depend on facilities, customers and supply chains, and await that engagement. The separation distance for ammonium nitrate in particular needs a quantity-distance basis that this document does not supply. The time held of Section 3.3, for material awaiting disposal, is also a parameter without a value. Pending a quantity-distance basis, the class records the references the Board gave: the Fire Protection Research Foundation's study of 2014, the United Kingdom's Health and Safety Executive, the Western Australian code's 300, 240 and 140 meters for more than 10 tonnes, and the one-mile default of NFPA 400 (*West Fertilizer Explosion* F-2). The element of encroachment, that the authority permitting building near a store holds the distance with the facility, is offered to the framework (Section 10.8).

10.2 Design-Basis Durations

The framework sets no durations (framework §4.2). The durations in Section 5 are class parameters, and the time a facility must hold its safe condition and keep delivering without outside support awaits the framework's values.

10.3 Agricultural Chemical Retailers

Agricultural chemical retailers holding ammonium nitrate or anhydrous ammonia are placed in this class on the author's decision of 5 October 2026. The Food and Agriculture framework now says so at its boundary with the Chemical sector and in its register. The item is closed.

10.4 Citation of the Shared Release Model, and the Explosion Addition

This document is the second to cite the shared release model at *Chemical Production and Synthesis* §4, and is owed notice when the model moves, under Project 9 OI-18. Section 4.2 adds detonation to the model for this class. The addition belongs in the model itself and should move there when the model is next revised. Three further elements go with it, from *West Fertilizer Explosion* F-4 and F-5: the combined chain in which a detonation initiates a nearby inhalation-hazard release; a source term for detonation that is the largest mass a single fire can involve, its yield uncertainty stated and not used to reduce it, including every mass on site a detonation could initiate; and an exposure stage that counts fragments, glazing, post-blast fire, the infrastructure reached and occupancy at the worst hour (Sections 3.3 and 4.2). They are offered to Chemical Production and Synthesis as the model's holder.

10.5 Local Sovereignty in the DBA-MA Sense

The protective-function property at Section 6.6 cites *Local Sovereignty* §3.2 directly. The requirement beside it, holding the safe condition with no outside power, communications or support, is the property the DBA-MA parent framework's local sovereignty principle describes. Whether that principle is FD-LS's construct at all is held pending Project 5's answer, and the requirement is written here in this document's own words until then. The item is registered under Project 9 OI-16.

10.6 Who Bears the Cost of the Stricter Supply Criteria

The stricter criteria a sole supplier must meet exist for its customers' benefit, and placing the obligation does not decide who pays for it. This is the same question Project 9 OI-17 asks for cooling water, applied to the forward supply rule. Whether OI-17 is widened to cover it is the author's to decide.

10.7 Reference Events and the Use Case

The two reference events proposed at Section 9.2 were conferred by the framework at its §9.1 as DBA-CHEM-RE-5 and DBA-CHEM-RE-6. The instrument for West, *The West Fertilizer Explosion, April 2013* (DBA-CHEM-RE-5), is written and, under the determination of 9 October 2026 that gives reference events their own project, is held by Project 11. This revision takes in

what it returns: Erratum 1 and Erratum 2 of Project 11's note of 10 October 2026 (batch 15), from its F-10 and F-8, at Sections 2.1, 2.4, 8.4, 9.2 and this section, and its recommendations R-1 to R-7, R-9, R-11 and R-12, at Sections 1.2, 1.3, 2.2, 3.3, 4.2, 5.1, 5.6, 6.2 to 6.4, 6.6 to 6.8, 7.4, 8.2, 8.3, 9.2, 10.1 and 10.4. The explosion at the port of Beirut (DBA-CHEM-RE-6) is withdrawn on the author's triage ruling of 9 October 2026 and its designator retired; its F-1 and F-2 are carried by *West Fertilizer Explosion* F-11 and F-12, its instrument is kept, withdrawn, in Project 9, and this document cites Beirut's findings only through West's. The use case for the supply-denial scenario is written as DBA-CHEM-UC-3 (*The Sole Disinfectant Supplier Taken Down*); what it returns to the class is not taken in at this revision.

10.8 Elements Noted to Others

Several elements of this revision belong to other instruments and are noted to them, not taken up here. To the framework: the encroachment element of *West Fertilizer Explosion* F-2, that the authority permitting building near a store holds the separation distance with the facility; the dependency of *West Fertilizer Explosion* F-12, that a state custodian's holder of a disposition decision lies outside the enterprise class, at framework §3.5 and §5; and the notes of *LockerGoga Attacks* R-3 and R-6, channels outward and notice to the authorities, at framework §2. To Chemical Enterprise Coordination and Control: the supplier's stewardship as a condition of sale (*West Fertilizer Explosion* F-9) and the dependency of *West Fertilizer Explosion* F-12. To Chemical Production and Synthesis: the supplier's stewardship (*West Fertilizer Explosion* F-9), and the elements of the shared model at Section 10.4 (*West Fertilizer Explosion* F-4, F-5). To Bulk Chemical Transport: the counting of material in transport custody awaiting unloading in the facility's source term, at its §1.2 (*West Fertilizer Explosion* F-5). To the Emergency Services series: the pre-incident plan, the stand-off decision and the public alert of Sections 4.2 and 5.1 (*West Fertilizer Explosion* F-3). To DBA-TRAN-ND-FC1 (*Nodes and Terminals*): material awaiting disposal at a node (*West Fertilizer Explosion* F-11), which that class records at its §4.4.

References

DBA-CHEM Series and Parent Framework

Chemical Sector Framework (DBA-CHEM_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

Chemical Production and Synthesis (DBA-CHEM-PR-FC1) and *Bulk Chemical Transport* (DBA-CHEM-TR-FC2). Eclectic Technologies, 2026. The shared release model, and the custody boundary at the transport end.

The West Fertilizer Explosion, April 2013 (DBA-CHEM-RE-5). Eclectic Technologies, 2026. The class's reference event, conferred at framework §9.1 and held by Project 11; cited by the short title *West Fertilizer Explosion* and finding or section number.

The LockerGoga Attacks, March 2019 (DBA-CHEM-RE-7). Eclectic Technologies, 2026. Held by Project 11; its F-3 is noted to Section 6.6, and cited by the short title *LockerGoga Attacks*.

Chemical Enterprise Coordination and Control (DBA-CHEM-CorpCo) and *The Sole Disinfectant Supplier Taken Down* (DBA-CHEM-UC-3). Eclectic Technologies, 2026. The coordination class, and the use case of the supply-denial scenario.

DBA-MA Parent Framework. Eclectic Technologies. Source of the four military-action characteristics and the design-basis-under-military-action methodology.

Defense in Depth (DBA-MA-DiD). Eclectic Technologies, 2026. Independent-protection disposition step and declared-residual form (§3.4, §3.5).

Other Series

Transportation Sector Framework (DBA-TRAN_Sector_Framework). Eclectic Technologies, 2026. The release-and-network boundary applied here to port storage.

Water Treatment Plant (DBA-W-FC1). Eclectic Technologies, 2026. The receiving end of chlorine and hypochlorite supply, and its own chemical-supply requirement.

Nodes and Terminals (DBA-TRAN-ND-FC1). Eclectic Technologies, 2026. Material awaiting disposal at a transportation node.

Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

Regulatory, Standards and Investigations

EPA Risk Management Program, 40 CFR Part 68; OSHA Process Safety Management, 29 CFR 1910.119; OSHA storage of ammonium nitrate, 29 CFR 1910.109(i); Emergency Planning and Community Right-to-Know Act; 49 CFR Parts 171 to 180; NFPA 704; NFPA 400 (2016 edition); Executive Order 13650, *Improving Chemical Facility Safety and Security* (1 August 2013), section 6(b).

U.S. Chemical Safety and Hazard Investigation Board. Investigation report 2013-02-I-TX, West Fertilizer Company fire and explosion, West, Texas, 17 April 2013, January 2016. Testimony of the Board's chairperson to the Senate Committee on Environment and Public Works, 27 June 2013, and statement of its supervisory investigator at Dallas, April 2014. Bureau of Alcohol, Tobacco, Firearms and Explosives, announcement of 11 May 2016, as the press reported it.

Explosion at the port of Beirut, 4 August 2020: Human Rights Watch, *"They Killed Us from the Inside": An Investigation into the August 4 Beirut Blast*, 3 August 2021, and Reuters' reporting of 2020, as *West Fertilizer Explosion* §1.3 cites them. The event was conferred as DBA-CHEM-RE-6 and is withdrawn, its designator retired.

Revision History

Version	Date	Author	Description
0.1 — DRAFT	October 2026	Tim Roxey	Initial issue. Third facility-class instrument in the DBA-CHEM series, typed physical under the author's determination of 30 September 2026. Covers terminals, repackagers, blenders and agricultural chemical retailers, the last on the author's decision of 5 October 2026. Establishes that either consequence chain may govern in this class, with supply loss as the governing military-action scenario; the reference configurations; five class design parameters including the supply-continuity margin; the non-adversarial event envelope with West, Texas and Beirut as proposed reference events; the Autonomous Safe Terminal Condition with its supply half; and the dual design basis, with inventory optimization at a sole supplier as the worked example. Second citer of the shared release model, to which it adds detonation (OI-18). Cites FD-LS directly and states the endurance requirement in its own words (OI-16). Raises whether OI-17 covers the forward supply rule. Written in plain language.
0.2	October 2026	Tim Roxey	Section 10.3 brought up to date: the Food and Agriculture framework, revised the same day, now places agricultural chemical retailers in this class. No requirement, event category, parameter or finding changed.
0.3 — DRAFT	October 2026	Tim Roxey	Takes in what the class's reference event returns, now held by Project 11: Erratum 1 (from *West Fertilizer Explosion* F-10) at Sections 2.4, 9.2 and 10.7 and Erratum 2 (from F-8) at Sections 2.1 and 8.4, of Project 11's batch 15 note, and *West Fertilizer Explosion* R-1 to R-7, R-9, R-11 and R-12, the last two being Beirut's findings carried into West, with the note of *LockerGoga Attacks* R-3 and R-6 at §6.6. The account of West gives the Board's figures for the quantity, the responders, the guidance and the half-mile count, records what the facility reported in 2012, and states the Bureau's determination that the fire was intentionally set as the Bureau's, with no finding (§2.4); the patchwork description is attributed to the chairperson's testimony and the supervisory investigator's statement (§2.1), and §8.4 states the regime as it applied at West, Executive Order 13650 section 6(b) as it directs, what followed, and that the class's requirements do not wait on a federal listing. Material awaiting disposal is in the class for Chain R whoever holds custody, with a port store as a fourth reference configuration and the time held as a parameter (§1.2, §1.3, §3.3, §8.3); the source term counts every mass a detonation could initiate, transport custody included, the separation distance is held by two parties, and occupancy is counted at the worst hour (§3.3); the exposure counts fragments, glazing, post-blast fire and infrastructure, knowing in advance is an exercised pre-incident plan, and a fourth condition adds the combined blast-and-toxic chain, hazards being ranked by consequence (§2.2, §4.2). Section 5.1 states what makes a store detonable in fire, the stand-off decision and the public alert, separation from an inhalation-hazard store, and detection and suppression independent against a set fire (§5.6); §6.2, §6.3 and §6.7 state the set fire whose delayed detonation reaches responders; §6.6 adds the means and channels of the supply half; §6.8 names three relaxations; §7.4 gives the

Version	Date	Author	Description
			disposition decision a single holder and a standing rule of removal; §9.2 names who holds a small retailer's design basis with it. West is stated as conferred at framework §9.1 as DBA-CHEM-RE-5, written and held by Project 11; Beirut (DBA-CHEM-RE-6) is recorded as withdrawn on the author's triage ruling of 9 October 2026, its designator retired and its F-1 and F-2 carried by West F-11 and F-12, and is cited as an illustration. DBA-CHEM-UC-3 is recorded as written, its returns not taken in. A new Section 10.8 notes elements to the framework, including LockerGoga R-3 and R-6 at its §2, and to Chemical Enterprise Coordination and Control, Chemical Production and Synthesis, Bulk Chemical Transport, the Emergency Services series and DBA-TRAN-ND-FC1. Tables 5-1 and 9-1 and the References are conformed. An independent check of the revision against the instrument, the errata and the instrument's predecessor in Project 11's archive raised 6 items: 4 taken before issue, and two referred to the author, whether the FD-BL framing of *West Fertilizer Explosion* F-3 enters the class beside that of F-12 at §7.4, and the spelling "per cent". Issued under framework v0.10. Revision-history entries are left as records.